

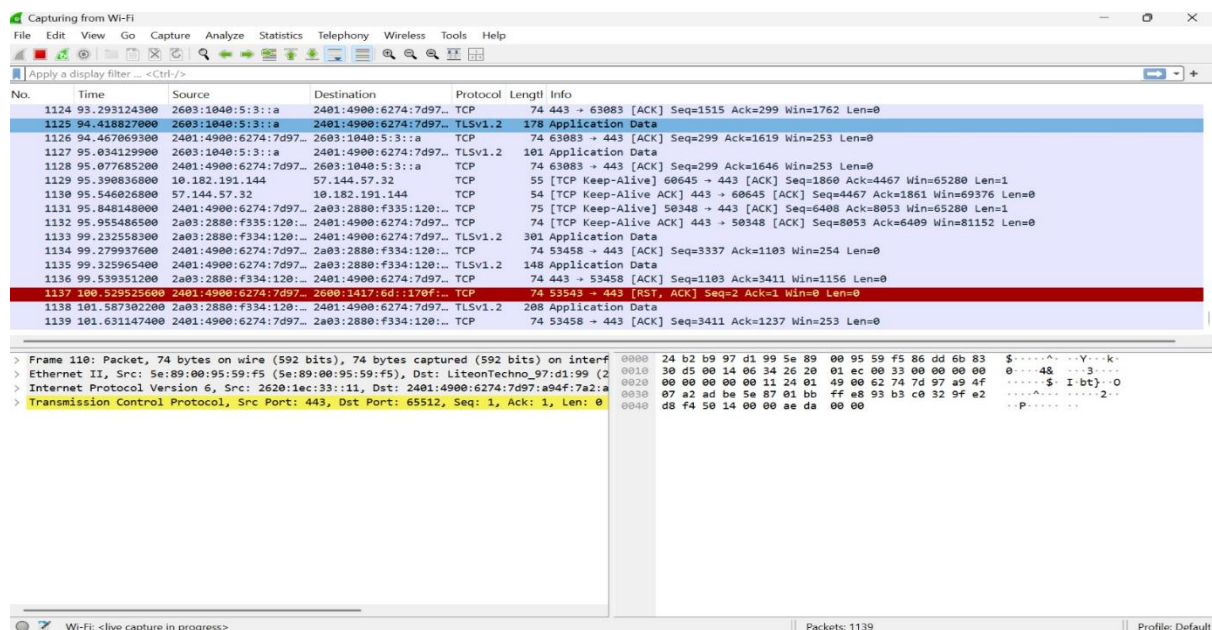
Password Capturing

Wireshark can capture not only passwords but any type of information transmitted over the network: usernames, email addresses, personal information, etc. As long as we can capture network traffic, Wireshark can sniff passing passwords.

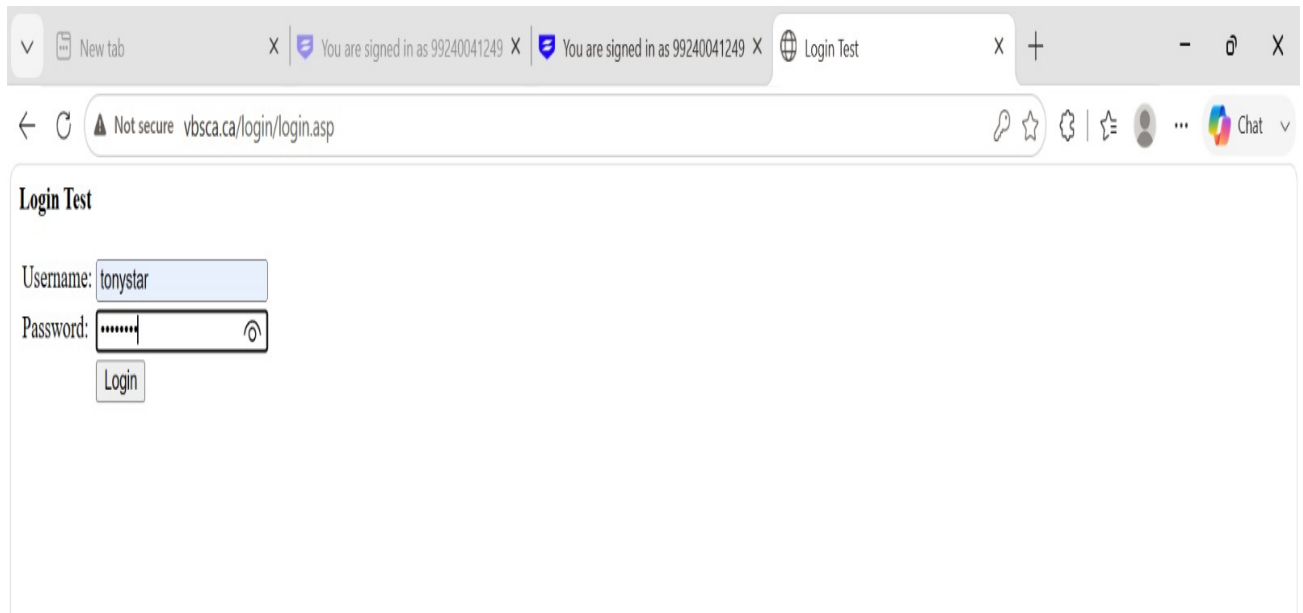
In sniffing can include passwords for various protocols such as HTTP, FTP, Telnet, etc. the captured data can be used to troubleshoot network problems, but can also be used maliciously to gain unauthorized access to sensitive information.

So, here we will see how we can capture the password using the Wireshark network capture analyzer. and see the outputs of the following steps.

Step 1: First of all, open your Wireshark tool in your window or in Linux virtual machine. and start capturing the network. suppose I am capturing my wireless fidelity.



Step 2: After starting the packet capturing we will go to the website and login the credential on that website as you can see in the image



Step 3: Now after completing the login credential we will go and capture the password in Wireshark. for that we have to use some filter that helps to find the login credential through the packet capturing.

Step 4: Wireshark has captured some packets but we specifically looking for HTTP packets. so in the display filter bar we use some command to find all the captured HTTP packets. as you can see in the below image the green bar where we apply the filter.

http

Wireshark packet capture showing HTTP traffic. The top pane displays a list of captured packets, with packet 51691 selected. The middle pane shows the details of the selected packet, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
28259	50.888341500	192.178.173.94	10.10.8.170	PKIX-C...	1107	Certificate Revocation List
25746	58.243075700	10.10.8.170	23.34.81.58	HTTP	165	GET /connecttest.txt HTTP/1.1
26479	58.959975300	23.34.81.58	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)
39082	89.992049000	10.10.8.170	23.34.81.48	HTTP	165	GET /connecttest.txt HTTP/1.1
39251	90.325084300	23.34.81.48	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)
51625	120.681741000	10.10.8.170	23.32.91.47	HTTP	165	GET /connecttest.txt HTTP/1.1
51691	120.981835600	10.10.8.170	23.155.129.172	HTTP	552	GET /login/login.asp HTTP/1.1
51720	121.002641400	23.32.91.47	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)
52096	121.626847900	23.155.129.172	10.10.8.170	HTTP	799	HTTP/1.1 200 OK (text/html)
52118	121.736166800	10.10.8.170	23.155.129.172	HTTP	510	GET /favicon.ico HTTP/1.1
52621	122.554123200	23.155.129.172	10.10.8.170	HTTP	1337	HTTP/1.1 404 Object Not Found (text/html)
57698	134.034836000	10.10.8.170	23.155.129.172	HTTP	785	POST /login/login_results.asp HTTP/1.1 (application/x-www-form-urlencoded)
57795	134.523175700	23.155.129.172	10.10.8.170	HTTP	143	HTTP/1.1 100 Continue
57920	134.870856100	23.155.129.172	10.10.8.170	HTTP	376	HTTP/1.1 200 OK (text/html)
64737	151.452240200	10.10.8.170	23.32.91.47	HTTP	165	GET /connecttest.txt HTTP/1.1
64855	151.664584200	23.32.91.47	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)

Frame 51691: Packet, 552 bytes on wire (4416 bits), 552 bytes captured (4416 bits) on Ethernet II, Src: LiteonTechno_97:d1:99 (24:b2:b9:97:d1:99), Dst: Sophos_fc:00:09 (c8:00:06:00:00:09), Seq: 1, Ack: 1, Len: 498

Hypertext Transfer Protocol

0030 00 ff 0c 45 00 00 47 45 54 20 2f 6c 6f 67 69 6e ... GET /login
0040 2f 6c 6f 67 69 6e 2e 61 73 70 20 48 54 5a 50 2f ... /login.asp HTTP/
0050 31 2e 31 0d 0a 48 6f 73 74 3a 20 76 62 73 63 61 ... 1.1. Host: vbcsa
0060 2e 63 61 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3aca. Connection:
0070 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 70 67 ... keep-alive
0080 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d 52 65 ... rade-Ins ecre-Re
0090 71 75 65 73 74 73 3a 20 31 0d 0a 55 73 65 72 2d ... quests: 1. User-
00a0 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 ... Agent: Mozilla/5
00b0 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54 20 31 ... 0 (Windows NT 1
00c0 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36 34 29 ... 0.0; Win 04; x64)
00d0 20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35 33 37 ... AppleWebKit/537
00e0 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69 6b 6536 (KHTML, like
00f0 20 47 65 63 6b 6f 29 20 43 68 72 6f 6d 65 2f 31 ... Gecko) Chrome/1
0100 35 31 2e 30 2e 30 2e 30 20 53 61 66 61 72 69 2f ... 51.0.0.0 Safari/
0110 35 33 37 2e 33 36 20 45 64 67 2f 31 35 31 2e 30 ... 537.36 Edge/151.0
0120 2e 30 2e 30 0d 0a 41 63 63 65 70 74 3a 20 74 650.0. Ac cept: te
0130 78 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 74 ... xt/html, applicat
0140 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70 ... ion/html level, ap
0150 70 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 30 ... plicatio n/xml;q=0
0160 30 2e 39 2c 69 6d 61 67 65 2f 61 76 69 66 2c 69 ... 0.9, imag e/avif,i
0170 6d 61 67 65 2f 77 65 62 70 2c 69 6d 61 67 65 2f ... mage/web p,image/
0180 61 70 6e 67 2c 2a 2f 2a 3b 71 3d 30 2e 38 2c 61 ... apng/*; q=0.8,a

Step 5: So there are some HTTP packets are captured but we specifically looking for form data that the user submitted to the website. for that, we have a separate filter

As we know that there are main two methods used for submitting form data from web pages like login forms to the server. the methods are-

GET

POST

Step 6: So firstly for knowing the credential we use the first method and apply the filter for the GET methods as you can see below.

`http.request.method == "GET"`

GET method

As you can see in the image there are two packets where the login page was requested with a GET request as well, but

The image shows a Wireshark packet capture of HTTP traffic. The packet list pane on the left shows several HTTP packets. Packet 51691 is highlighted, showing a GET request for /login/login.asp. The packet details pane on the right shows the structure of the HTTP request, including the request line, headers, and body. The request line is GET /login/login.asp HTTP/1.1. The headers include Host: vbbsca.ca, Connection: keep-alive, Upgrade-Insecure-Requests: 1, User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36, and Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8. The body is empty.

No.	Time	Source	Destination	Protocol	Length	Info
26479	58.959975300	23.34.81.58	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)
39082	89.992049000	10.10.8.170	23.34.81.48	HTTP	165	GET /connecttest.txt HTTP/1.1
39251	90.325084300	23.34.81.48	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)
51625	120.681741000	10.10.8.170	23.32.91.47	HTTP	165	GET /connecttest.txt HTTP/1.1
51691	120.901835600	10.10.8.170	23.155.129.172	HTTP	552	GET /login/login.asp HTTP/1.1
51720	121.002641400	23.32.91.47	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)
52096	121.626847900	23.155.129.172	10.10.8.170	HTTP	799	HTTP/1.1 200 OK (text/html)
52110	121.736166800	10.10.8.170	23.155.129.172	HTTP	510	GET /favicon.ico HTTP/1.1
52621	122.554123200	23.155.129.172	10.10.8.170	HTTP	1337	HTTP/1.1 404 Object Not Found (text/html)
57698	134.034836000	10.10.8.170	23.155.129.172	HTTP	785	POST /login/login_results.asp HTTP/1.1 (application/x-www-form-urlencoded)
57795	134.523175700	23.155.129.172	10.10.8.170	HTTP	143	HTTP/1.1 100 Continue
57920	134.878056100	23.155.129.172	10.10.8.170	HTTP	376	HTTP/1.1 200 OK (text/html)
64737	151.452240200	10.10.8.170	23.32.91.47	HTTP	165	GET /connecttest.txt HTTP/1.1
64855	151.664584200	23.32.91.47	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)
78875	183.831610600	10.10.8.170	23.34.81.42	HTTP	165	GET /connecttest.txt HTTP/1.1
79016	184.351301400	23.34.81.42	10.10.8.170	HTTP	241	HTTP/1.1 200 OK (text/plain)

Frame 51691: Packet, 552 bytes on wire (4416 bits), 552 bytes captured (4416 bits) on Ethernet II, Src: LiteonTechno_97:d1:99 (24:b2:b9:97:d1:99), Dst: Sophos_fc:00:09 (08:00:0c:00:00:09), Internet Protocol Version 4, Src: 10.10.8.170, Dst: 23.155.129.172, Transmission Control Protocol, Src Port: 64484, Dst Port: 80, Seq: 1, Ack: 1, Len: 552, Hypertext Transfer Protocol

GET /login/login.asp HTTP/1.1

Request Method: GET
Request URI: /login/login.asp
Request Version: HTTP/1.1
Host: vbbsca.ca
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Referer: https://www.google.com/
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9,en-IN;q=0.8

there is no form data submitted with a GET request.

Step 7: Now after checking the GET method if we didn't find the form data, then we will try the POST method for that we will apply the filter on Wireshark as you can see.

http.request.method == "POST"

The image shows a Wireshark packet capture of HTTP traffic. The packet list pane on the left shows several HTTP packets. Packet 75953 is highlighted, showing a POST request for /login/login_results.asp. The packet details pane on the right shows the structure of the HTTP request, including the request line, headers, and body. The request line is POST /login/login_results.asp HTTP/1.1. The headers include Host: vbbsca.ca, Connection: keep-alive, Upgrade-Insecure-Requests: 1, User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36, and Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8. The body contains form data: txtUsername=tonystar and txtPassword=12345678.

No.	Time	Source	Destination	Protocol	Length	Info
71165	523.728497100	2600:1417:6e::170f::...	2401:4900:6274:7d97::...	HTTP	261	HTTP/1.1 200 OK (text/plain)
71550	593.914152300	2401:4900:6274:7d97::...	2600:1417:6e::170f::...	HTTP	222	GET /connecttest.txt HTTP/1.1
71552	593.914416300	10.182.191.144	23.205.118.154	HTTP	201	GET /connecttest.txt HTTP/1.1
71553	594.323023600	23.205.118.154	10.182.191.144	HTTP	241	HTTP/1.1 200 OK (text/plain)
71557	594.425450500	2600:1417:6e::170f::...	2401:4900:6274:7d97::...	HTTP	261	HTTP/1.1 200 OK (text/plain)
71763	629.038462500	10.182.191.144	23.205.118.154	HTTP	201	GET /connecttest.txt HTTP/1.1
71764	629.038517700	2401:4900:6274:7d97::...	2600:140f:2400::686::...	HTTP	222	GET /connecttest.txt HTTP/1.1
71766	629.513009600	2600:140f:2400::686::...	2401:4900:6274:7d97::...	HTTP	261	HTTP/1.1 200 OK (text/plain)
71768	629.513179000	23.205.118.154	10.182.191.144	HTTP	241	HTTP/1.1 200 OK (text/plain)
75452	731.997994400	10.182.191.144	23.155.129.172	HTTP	550	GET /login/login.asp HTTP/1.1
75487	732.855011600	23.155.129.172	10.182.191.144	HTTP	799	HTTP/1.1 200 OK (text/html)
75496	732.950079700	10.182.191.144	23.155.129.172	HTTP	510	GET /favicon.ico HTTP/1.1
75589	734.402407000	23.155.129.172	10.182.191.144	HTTP	57	HTTP/1.1 404 Object Not Found (text/html)
75953	752.861832900	10.182.191.144	23.155.129.172	HTTP	785	POST /login/login_results.asp HTTP/1.1 (application/x-www-form-urlencoded)
76049	753.353978100	23.155.129.172	10.182.191.144	HTTP	143	HTTP/1.1 100 Continue
76058	753.982406000	23.155.129.172	10.182.191.144	HTTP	376	HTTP/1.1 200 OK (text/html)

Frame 75953: Packet, 785 bytes on wire (6280 bits), 785 bytes captured (6280 bits) on Ethernet II, Src: LiteonTechno_97:d1:99 (24:b2:b9:97:d1:99), Dst: 5e:89:00:95:59:f5 (50:00:00:00:00:00), Internet Protocol Version 4, Src: 10.182.191.144, Dst: 23.155.129.172, Transmission Control Protocol, Src Port: 57374, Dst Port: 80, Seq: 1, Ack: 1, Len: 731, Hypertext Transfer Protocol

POST /login/login_results.asp HTTP/1.1

Host: vbbsca.ca
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Referer: https://www.google.com/
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9,en-IN;q=0.8

Form item: "txtUsername" = "tonystar"
Form item: "txtPassword" = "12345678"

As you can see we have a packet with form data click on the packet with user info and the application URL encoded. and click on the down-

HTML form URL Encoded where the login credential is found. login credential as it is the same that we filed on the website in step 2.

Form item: "uname" = "tonystar"

Form item: "pass" = "12345678"